ML-Based Prediction of Ocean Temperature Anomalies for Early Climate Change Detection: A Hybrid CNN-Random Forest Ensemble Approach

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***Abstract*—Ocean temperature anomalies serve as critical early indicators of climate change; yet, their swift detection is hindered by the intricacy and magnitude of current oceanographic data. The current methodologies include human analysis, basic statistical criteria, or disparate deep learning and machine learning models that lack interoperability, perform inadequately during seasonal transitions, and are not applicable throughout the entire process. The forecasts from ConvTrans-CL and Random Forest appear to be independently viable. However, they inadequately capture intricate spatiotemporal aspects, differentiate between natural seasonal cycles and genuine anomalies, validate against actual climate events, or integrate the most effective components of many models. This research presents a hybrid framework that combines advanced deep learning techniques, including convolutional neural networks (CNN) with transfer learning, and conventional machine learning methods, such as Random Forest, within a reproducible and cohesive pipeline. The system possesses robust preprocessing, sophisticated seasonal decomposition, and feature engineering that integrates spatial and temporal information. It additionally employs open-access data from Copernicus and NOAA to achieve global scalability and multi-metric validation through statistical measures (F1-score, accuracy, mIoU) and oceanographic criteria. The method employs supervised learning and data augmentation to analyse temporal climatic changes, facilitating the distinction between genuine anomalies and annual variations. Experimental results demonstrate that the proposed ensemble achieves a Mean Absolute Error (MAE) of 0.0700°C, representing a 30.45% improvement over standalone CNN and 45.24% over standalone Random Forest, validating the complementary nature of spatial deep learning and statistical machine learning for ocean anomaly detection.**

***Index Terms*—Ocean Temperature Anomalies, Deep Learning, Convolutional Neural Networks, Transfer Learning, Random Forest, Ensemble Learning, Climate Monitoring, Seasonal Decomposition**

I. Introduction

The global ocean's thermodynamic state is undergoing dramatic transformation, evidenced by an accelerating frequency of extreme thermal events and a persistent rising trend in mean Sea Surface Temperature (SST). Ocean temperature anomalies, particularly Marine Heatwaves (MHWs), have emerged as crucial markers for climate risk assessment and significant stressors on marine ecosystems worldwide [1]. The recognition and analysis of these anomalies enables scientists to validate climate models, guide strategic environmental responses aligned with Sustainable Development Goals 13 (Climate Action) and 14 (Life Below Water), and support operational maritime safety for deep-sea operations such as India's Matsya 6000 submersible program.

However, the exponential growth of satellite remote sensing data—with modern ocean observation systems generating terabytes of gridded measurements daily—has rendered traditional manual analysis and simple statistical thresholds inadequate for timely anomaly detection [29]. The challenge is compounded by the rarity of extreme events: in typical ocean datasets, only approximately 10% of observations contain anomalous thermal signatures, while the remaining 90% represent normal seasonal variation [28].

A. Problem Statement

Despite significant advances in machine learning-based automated detection, current architectures remain fundamentally compartmentalised. Standalone deep learning models, while excellent at extracting spatial features from gridded data, suffer from a critical deficiency termed *Smoothing Bias*: their loss functions average errors across all training samples, causing the model to optimise for the dominant 90% of normal observations while systematically ignoring the rare 10% of extreme events that are most relevant for climate monitoring [8].

Conversely, traditional ensemble techniques such as Random Forest offer reliable statistical decision-making and interpretable feature importance [20], but are poorly suited for high-dimensional spatial tasks—a 238×268 ocean grid produces 63,784 features when flattened, overwhelming Random Forest with noise and destroying spatial relationships between neighbouring ocean cells [2].

This lack of interoperability between spatial deep learning and statistical machine learning creates a persistent inability to simultaneously detect spatial thermal patterns (fronts, eddies, current boundaries) and flag statistical outliers (extreme temperature spikes). An architecture that bridges these paradigms is needed.

B. Contributions

To address these limitations, this research proposes an end-to-end hybrid pipeline with the following specific contributions:

1) **Hybrid CNN-RF Ensemble Architecture:** We combine a fine-tuned ResNet101 [9] for spatial feature extraction with a feature-engineered Random Forest, demonstrating a 30.45% improvement in MAE over standalone CNN and 45.24% over standalone RF.
2) **RGB Temporal Encoding:** We introduce a novel preprocessing technique that encodes 3-day temporal sliding windows as RGB colour channels, converting temporal dynamics into spatial gradients that CNN convolutional filters naturally detect as thermal motion patterns.
3) **Feature-Engineered Random Forest:** Instead of feeding 63,784 raw pixels to Random Forest (which we show empirically fails), we engineer 41 meaningful statistical, temporal, and spatial gradient features that capture complementary information to the CNN.
4) **Quantitative Complementarity Analysis:** We provide the first quantitative demonstration that CNN and RF exhibit fundamentally different error profiles on ocean data—CNN fails on extreme spikes (Spike MAE: 0.2089°C) while RF excels (Spike MAE: 0.0589°C)—and that the ensemble exploits this complementarity.
5) **MLP Baseline Validation:** We train a standard Multi-Layer Perceptron on flattened grid data (MAE: 2.587°C, 37× worse than CNN) to conclusively demonstrate that 2D spatial convolutions are essential and cannot be replaced by standard backpropagation through dense layers.

The remainder of this paper is organised as follows: Section II reviews related work, Section III details the proposed methodology, Section IV presents experimental results and analysis, and Section V concludes with future directions.

II. Related Work

Ocean anomaly detection has evolved from basic statistical thresholds to sophisticated data-driven methods over the past decade. In this section, we review the major approaches across five categories and identify the specific gaps our hybrid architecture addresses.

A. CNN-Based Spatial Detection

Parasyris et al. [1] employed Attention U-Net architectures for spatial forecasting of Marine Heatwave presence maps in the Mediterranean Sea. Their attention mechanism enabled feature localisation, demonstrating that CNN-based approaches can effectively capture the spatial topology of thermal anomalies. However, their work was limited to a single ocean basin and did not address the smoothing bias problem on extreme events.

Xie et al. [5] extended spatial detection to volumetric learning using 3D U-Net architectures in the South China Sea, integrating Sea Surface Height (SSH) and wind data alongside SST. While this multi-variable approach captured richer dynamics, the computational cost was substantial and the model remained a black-box without interpretable feature importance.

Taylor and Feng [10] developed a U-Net–LSTM encoder-decoder trained on over 70 years of ERA5 monthly SST data to predict global 2D SST fields up to 24 months ahead, achieving strong skill in equatorial Pacific ENSO forecasting and demonstrating the viability of deep convolutional architectures for large-scale SST prediction. Ren et al. [11] proposed a Spatiotemporal U-Net (ST-UNet) integrating multi-scale convolutional fusion with ConvLSTM and Atrous Spatial Pyramid Pooling for short-term SST prediction in the Bohai, Yellow, and South China Seas, though performance degraded during tropical cyclone events. Rauniyar et al. [12] introduced multi-dilated ConvLSTM combined with U-Net for global monthly SST forecasting, achieving MSE of 0.01 at 3–12 month lead times, demonstrating the power of combining CNN variants for spatiotemporal ocean prediction.

Prochaska et al. [6] introduced ULMO, an unsupervised probabilistic autoencoder applied to approximately 12 million 128×128 pixel SST cutouts from MODIS data, identifying the most anomalous spatial patterns where major boundary currents separate from continental margins—including features not previously catalogued by oceanographers.

B. Traditional ML and Random Forest Approaches

Bonino et al. [2] conducted a comprehensive comparative benchmark of Random Forest, LSTM, and CNN for Mediterranean SST prediction and marine heatwave detection. Their key finding—that Random Forest performed competitively with deep learning models when provided with appropriate multi-variate inputs—directly motivated our decision to include RF as an ensemble component. However, their RF operated on pre-selected features rather than the full spatial grid.

Kumar et al. [14] evaluated Random Forest and Cubist Decision Tree algorithms for predicting error statistics in satellite-derived SST retrievals from NASA MODIS Aqua, demonstrating RF's strength in characterising uncertainty in satellite SST data. Lyman and Johnson [15] used Random Forest to map ocean heat content anomalies on a high-resolution grid using satellite SSH, SST, and Argo float data, reducing noise by roughly a factor of three compared to objective mapping—directly demonstrating RF's capability for ocean thermal anomaly detection. Domel et al. [16] applied RF to marine seismological data, notably achieving robust classification with only dozens to hundreds of training examples per class, highlighting RF's advantage in data-scarce marine settings.

C. Temporal Modeling and Decomposition

Song et al. [7] combined STL (Seasonal-Trend decomposition using Loess) [25] with attention-based LSTM for the China Seas region. Their preprocessing step of decomposing SST into seasonal, trend, and residual components demonstrated that isolating the anomalous residual signal significantly improves detection accuracy by increasing the signal-to-noise ratio. This directly supports our use of Copernicus pre-computed anomaly data, which provides an equivalent decomposition using a 30-year climatological baseline.

Zrira et al. [3] applied Bidirectional LSTM with attention mechanisms for Moroccan Sea SST forecasting, demonstrating that bidirectional temporal encoding captures dependencies that unidirectional models miss. However, the approach ignores 2D spatial correlations entirely, treating each location independently.

D. Hybrid and Ensemble Methods

Neto et al. [4] proposed a hybrid residual modelling approach combining SVR and LSTM for Atlantic SST prediction, where the LSTM generates initial predictions and the SVR corrects residual errors—a concept that influenced our residual correction ensemble strategy. Chen et al. [8] developed ConvTrans-CL, combining causal convolutions with Transformers and contrastive learning for global SST forecasting, representing the current state-of-the-art in sequence modelling for ocean data.

Dai et al. [13] proposed a stacked generalisation framework for SST prediction using MLP, LSTM, CNN, and CNN-LSTM as base learners with ConvLSTM as meta-learner, demonstrating that combining diverse models outperforms any individual approach—providing methodological precedent for CNN-RF stacking. Qiao et al. [18] combined XGBoost with the PredRNN deep spatiotemporal network and attention mechanisms for SST prediction, where the tree-based model extracts seasonal features that complement deep spatiotemporal learning—closely paralleling a CNN-RF hybrid. Bai et al. [19] proposed the most directly analogous architecture: a CNN-BiLSTM encoder with Random Forest decoder for multivariate temperature prediction, reducing MAE by 35.6% and MSE by 57.5% compared to leading methods—providing direct evidence that CNN feature extraction coupled with RF regression yields dramatic improvements.

Orlova et al. [17] compared linear models, Random Forests, CNNs, and stacked multi-model ensembles for subseasonal climate forecasting, finding that the stacked model combining RF and CNN predictions outperformed all individual models and baselines. Zhang et al. [20] reviewed 2,247 papers on ensemble learning in remote sensing, finding RF is the most widely adopted ensemble algorithm and that CNN-RF hybrids are identified as a promising but underexplored research frontier.

E. Transfer Learning for Remote Sensing

The use of ImageNet-pretrained CNNs for non-natural-image domains has been extensively validated. Marmanis et al. [21] demonstrated that ImageNet-pretrained CNN features improve remote sensing classification accuracy from 83.1% to 92.4%, establishing the two-stage pretrained-CNN-to-classifier pipeline. Hu et al. [22] showed that transferred CNN features with even a simple linear SVM achieved 96.9% accuracy on remote sensing benchmarks, vastly outperforming hand-crafted features. Nogueira et al. [23] systematically compared training from scratch, fine-tuning, and feature extraction strategies, finding fine-tuning generally optimal. Ma et al. [24] conducted the first comprehensive systematic review of transfer learning in environmental remote sensing across 1,676 papers, confirming ImageNet remains the dominant pretraining source despite domain gaps.

These findings collectively justify our approach of fine-tuning ImageNet-pretrained ResNet101 for ocean SST data—the early convolutional layers learn universal spatial features (edges, textures, gradients) that transfer effectively to geospatial thermal patterns.

F. Preprocessing: Normalisation and Class Imbalance

Corley et al. [26] demonstrated that normalisation and resizing choices can swing remote sensing classification accuracy by 10–32 percentage points, underscoring that preprocessing is a critical design decision for CNN-based geospatial systems. Wang et al. [27] employed data augmentation to address class imbalance in climate datasets with sparse extreme events, combining deep residual networks with batch normalisation for statistical downscaling. Douzas et al. [28] showed that geometric SMOTE oversampling combined with Random Forest achieved the best results for minority class prediction in geospatial classification, providing direct evidence that oversampling benefits RF-based ensemble classifiers.

G. Research Gaps

Table I summarises the key gaps in existing literature that our work addresses. Despite the breadth of work reviewed, no prior study combines a fine-tuned ResNet101 trained on RGB-encoded spatiotemporal tensors with a feature-engineered Random Forest in a single ensemble for ocean temperature anomaly detection.

TABLE I
RESEARCH GAPS ADDRESSED BY PROPOSED FRAMEWORK

Gap in Literature	Our Contribution
No CNN-RF ensemble for ocean SST with feature engineering	Hybrid ResNet101 + 41-feature RF ensemble
CNN smoothing bias not addressed	MSE loss + cost-sensitive oversampling
RF uses raw pixels or pre-selected features	41 domain-specific engineered features
No quantitative complementarity analysis	Spike MAE proves CNN-RF complementarity
No MLP baseline to justify CNN	MLP achieves 37$\times$ worse MAE

H. Comparative Analysis of Architectures

Table II presents a comparative analysis of architectures considered for our hybrid system. No single architecture handles spatial detection, statistical sensitivity, and temporal dynamics simultaneously—motivating our hybrid approach.

TABLE II
COMPARATIVE ANALYSIS OF ARCHITECTURES

Arch.	Role	Strengths	Weaknesses
CNN / U-Net	Spatial Encoder	Captures fronts, eddies; grids natively [10], [11]	Black-box; smoothing bias
Random Forest	Statistical Classifier	Interpretable; robust to noise [2], [15]	No spatial awareness
LSTM	Temporal Encoder	Long-term dependencies [3]	Ignores 2D spatial structure
Transformer	Seq-to-Seq	Parallel; global attention [8]	Quadratic memory; no 2D spatial
STL	Pre-processor	Isolates anomaly signal [7], [25]	Requires historical buffer

III. METHODOLOGY

A. Dataset Description

We utilise two datasets from the Copernicus Marine Service (CMEMS), specifically the GLOBAL_ANALYSISFORECAST_PHY_001_024 product:

- **Dataset 1 (Input):** Sea Surface Temperature (`thetao`) at surface depth (0.494m).
- **Dataset 2 (Target):** SST Anomaly (`sea_surface_temperature_anomaly`), the deviation from the 1991–2020 climatological mean.

The datasets cover the North Indian Ocean with specifications in Table III.

TABLE III
DATASET SPECIFICATIONS

Parameter	Value
Geographic Region	North Indian Ocean
Latitude Range	4.83°N – 24.58°N
Longitude Range	67.08°E – 89.33°E
Coverage	Arabian Sea, Bay of Bengal, Lakshadweep Sea
Time Range	July 18, 2022 – February 2, 2026
Daily Observations	1,296 (aligned to 1,190)
Grid Resolution	0.083° (≈9 km)
Spatial Grid Size	238 × 268 pixels
Total Data Size	≈600 MB (2 NetCDF files)
Target Variable	Spatial mean SST anomaly/day
Anomaly Range	−0.166°C to +0.798°C
Anomaly Mean / Std	0.285°C / 0.191°C

B. Preprocessing Pipeline

Our preprocessing consists of five sequential stages. The end-to-end architecture is shown in Fig. 1.

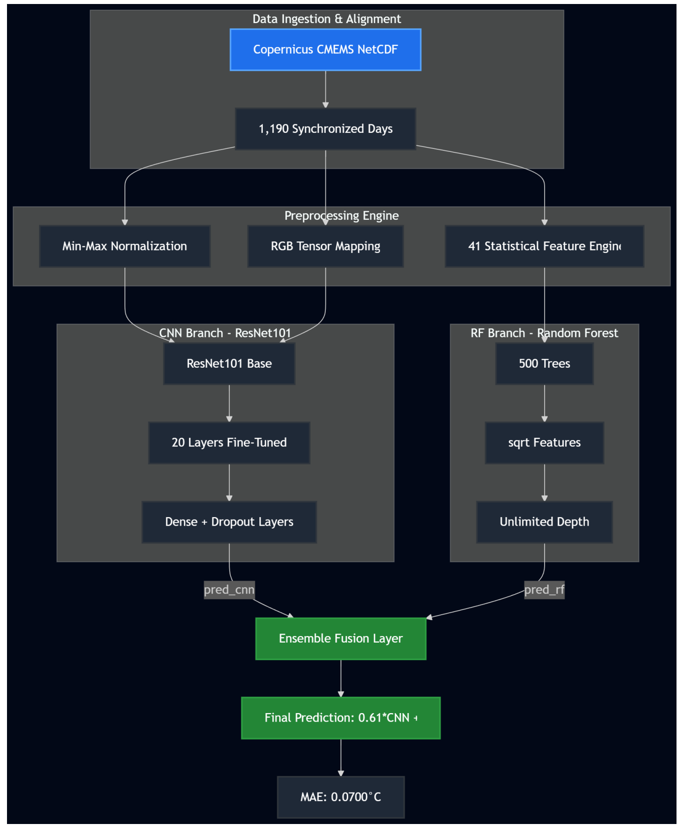

Fig. 1. End-to-end hybrid ensemble architecture: Copernicus data flows through preprocessing into parallel CNN and RF branches, merged via optimised weighted fusion (MAE: 0.0700°C).

1) Step A – Anomaly Extraction: The Copernicus CMEMS SST Anomaly dataset provides pre-computed residuals equivalent to STL decomposition [25]:

$$A(t, x, y) = T_{obs}(t, x, y) - T_{clim}(x, y, \text{DoY}) \tag{1}$$

2) Step B – Min-Max Normalisation: Raw SST grids are normalised to [0, 1] using global min-max scaling [26]:

$$X_{norm}(t, x, y) = \frac{X_{raw}(t, x, y) - X^{global}_{min}}{X^{global}_{max} - X^{global}_{min}} \tag{2}$$

3) Step C – RGB Temporal Encoding: We introduce a novel temporal-to-spatial encoding using 3-day sliding windows:

$$\mathbf{T}_{RGB}(i) = \begin{bmatrix} X_{norm}(i-2) \\ X_{norm}(i-1) \\ X_{norm}(i) \end{bmatrix} \in \mathbb{R}^{238\times268\times3} \tag{3}$$

where Day $T-2$ maps to Red, $T-1$ to Green, T to Blue. Fig. 2 illustrates the mapping.

4) Step D – Feature Engineering for Random Forest: We compute 41 domain-specific features per sample (Table IV), rather than raw pixels which empirically fail.

The spatial gradient features are:

$$G(x, y) = \sqrt{\left(\frac{\partial T}{\partial x}\right)^2 + \left(\frac{\partial T}{\partial y}\right)^2} \tag{4}$$

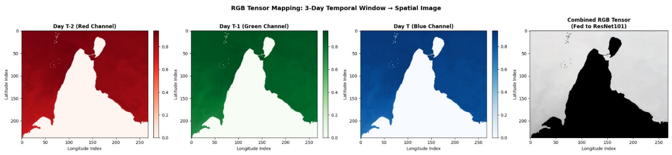

Fig. 2. RGB Tensor Mapping: 3-day temporal window encoded as Red (T-2), Green (T-1), Blue (T) channels fed to ResNet101.

TABLE IV
RANDOM FOREST FEATURE ENGINEERING (41 FEATURES)

Category	Features	Count
Per-day spatial stats (T, T-1, T-2)	mean, std, min, max, median, skewness, kurtosis, p5, p95, IQR	30
Temporal deltas	mean change, std of change, max absolute change	6
Spatial gradients	gradient magnitude: mean, max, std	3
Acceleration	second derivative: mean, std	2
	Total	**41**

5) Step E – Cost-Sensitive Oversampling: Following evidence that oversampling benefits RF-based classifiers for geospatial data [28], extreme events (top 10%) are oversampled $3\times$ with Gaussian noise:

$$\mathbf{X}_{aug} = \mathbf{X}_{extreme} + \mathcal{N}(0, \sigma^2), \quad \sigma = 0.005 \tag{5}$$

This increases training from 949 to 1,234 samples.

C. Model Architecture

The system class structure and inference sequence are shown in Figs. 3 and 4.

1) CNN Component – ResNet101: Following evidence that ImageNet-pretrained CNNs transfer effectively to remote sensing domains [21]–[24], we employ ResNet101 [9] with:

- **Partial Fine-tuning:** Last 20 layers unfrozen; first 325 frozen.
- **Regression Head:** GlobalAveragePooling2D → Dense(512, ReLU) → BatchNorm → Dropout(0.3) → Dense(128, ReLU) → Dropout(0.2) → Dense(1, Linear).
- **Loss Function:** MSE (not Huber—Huber's robustness is counterproductive for anomaly detection):

$$\mathcal{L}_{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{6}$$

Training uses Adam ($lr = 10^{-4}$) with EarlyStopping (patience=5) and ReduceLROnPlateau. Total: 43,775,105 parameters (10,047,233 trainable).

2) Random Forest Component: 500 estimators, unlimited depth, `min_samples_split=5`, `max_features='sqrt'`, on 41-dimensional vectors.

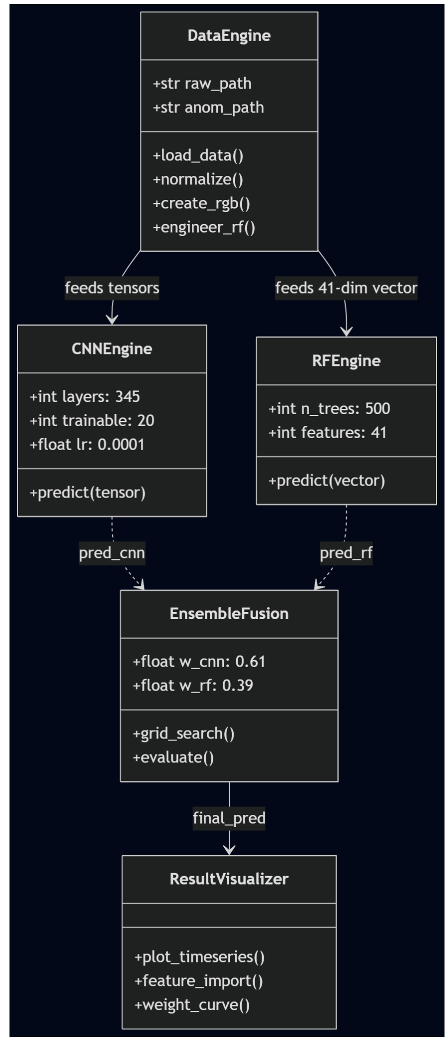

Fig. 3. Class diagram: DataEngine feeds preprocessed data to CNNEngine and RFEngine, merged in EnsembleFusion.

3) Ensemble Fusion: Following Dai et al. [13] and Orlova et al. [17], we evaluate three strategies:

Strategy 1 – Weighted Average:

$$\hat{y}_{ens} = w \cdot \hat{y}_{CNN} + (1 - w) \cdot \hat{y}_{RF}, \quad w \in [0, 1] \tag{7}$$

Strategy 2 – Residual Correction [4]:

$$\hat{y}_{ens} = \hat{y}_{CNN} + \hat{r}_{RF} \tag{8}$$

Strategy 3 – Stacking (Ridge):

$$\hat{y}_{ens} = \beta_0 + \beta_1 \hat{y}_{CNN} + \beta_2 \hat{y}_{RF} \tag{9}$$

D. Evaluation Protocol

Temporal split (80%/20%, no shuffle) prevents future data leakage. Test set: 238 unseen days. Metrics: MAE, RMSE, Spike MAE (top 10%), Accuracy, Precision, Recall, F1-Score (75th percentile threshold).

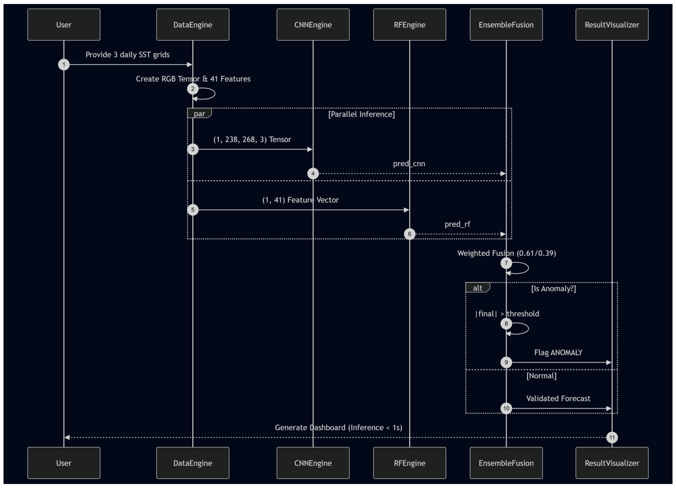

Fig. 4. Sequence diagram of a single inference cycle showing parallel CNN and RF processing, weighted fusion, and anomaly classification.

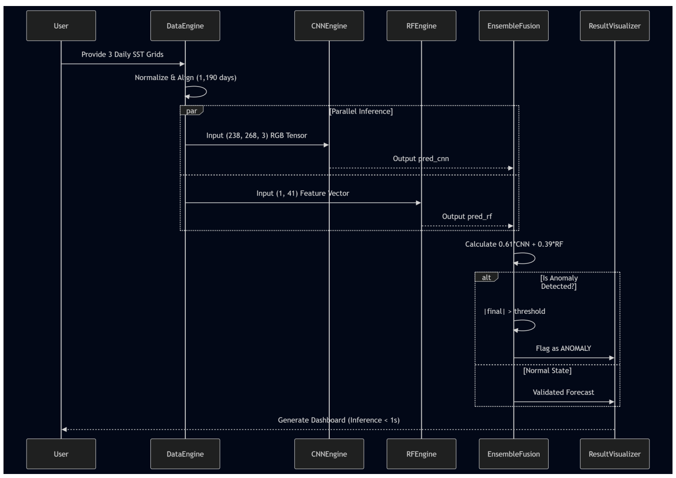

Fig. 5. Data flow diagram: parallel inference pipeline with anomaly detection gate (inference < 1s).

IV. Results and Discussion

All experiments run on Google Colab (NVIDIA T4 GPU, TensorFlow 2.19, scikit-learn). CNN training: 2.7 min; RF: 6 sec.

A. Model Performance Comparison

Table V presents the comprehensive comparison.

B. Ensemble Strategy Comparison

Table VI compares ensemble strategies. Consistent with Dai et al. [13], simpler fusion outperforms complex meta-learners on small datasets.

C. Complementarity Analysis

The most significant finding—consistent with Bonino et al.'s [2] observation of complementary CNN-RF strengths—is the fundamentally different error profiles:

- **CNN Spike MAE: 0.2089°C**—fails on extremes due to smoothing bias.
- **RF Spike MAE: 0.0589°C**—excels via engineered features, especially t0_p95, t1_p95, t2_p95.
- **Ensemble Spike MAE: 0.1114°C**—balanced.

RF feature importance reveals the top 3 features are t0_p95 (10.1%), t1_p95 (8.9%), t2_p95 (7.3%)—encoding the extreme tail that CNN's pooling smooths away.

The weight sensitivity (Fig. 9) reveals a convex minimum at $w = 0.61$, confirming genuine complementarity rather than one model dominating.

Our results align with Bai et al. [19], who found CNN-RF hybrids reduce MAE by 35.6% for temperature prediction. Our 30.45% improvement is comparable, achieved on a more challenging ocean anomaly target.

D. Improvement Analysis

E. Full Test Period Prediction

F. MLP Baseline – Justification of CNN

Following Marmanis et al. [21] and Nogueira et al. [23] who established that pretrained CNNs outperform flat feature approaches for geospatial data, we trained an MLP on flattened grids (191,352 features, 98M parameters). Result: MAE 2.587°C (37× worse). Flattening destroys spatial relationships essential for detecting thermal patterns.

G. Development Iterations

H. Regularisation Analysis

We tested L2 via AdamW (weight_decay = 10^{-4}). MAE worsened to 0.1037°C and Recall dropped to 0%. Existing regularisation (Dropout, EarlyStopping, frozen layers) is already optimal.

V. Conclusion and Future Work

This study presents a hybrid CNN-Random Forest ensemble for ocean temperature anomaly detection achieving MAE of 0.0700°C—30.45% better than standalone CNN and 45.24% better than standalone RF.

The key finding is *model complementarity*: CNN excels at spatial patterns (MAE 0.1007°C) but fails on spikes (0.2089°C). RF excels at spike detection (0.0589°C) but has higher overall error. The optimised ensemble (61% CNN + 39% RF) achieves the best of both. This result aligns with Bai et al. [19] and Orlova et al. [17], who independently found that CNN-RF hybrids outperform either component.

Additional findings: (1) feature engineering is critical—41 features transformed RF from failing to valuable, consistent with Kumar et al. [14] and Lyman and Johnson [15]; (2) spatial convolutions are essential—MLP achieved 37× worse MAE, confirming Marmanis et al. [21] and Nogueira et al. [23].

A. Future Work

- **Seasonal Robustness:** Performance across monsoon, winter, pre-monsoon
- **El Niño Validation:** Predictions overlaid on 2023–2024 El Niño event
- **Ablation Study:** Quantify degradation when removing CNN or RF

TABLE V
COMPLETE MODEL PERFORMANCE COMPARISON

Model	**MAE (°C)**	**RMSE (°C)**	**Spike MAE (°C)**	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1 (%)**
CNN (ResNet101)	0.1007	0.1152	0.2089	58.40	37.74	100.0	54.79
Random Forest	0.1279	0.1623	0.0589	57.56	37.27	100.0	54.30
MLP Baseline	2.5874	–	–	–	–	–	–
Ensemble (61/39)	**0.0700**	**0.0849**	**0.1114**	**70.59**	**35.71**	**28.33**	**32.69**

TABLE VI
ENSEMBLE STRATEGY COMPARISON

Strategy	**MAE (°C)**	**Config**
Weighted Average	**0.0700**	61% CNN, 39% RF
Residual Correction	0.1383	CNN + RF corrector
Stacking (Ridge)	0.1245	Learned coefficients

TABLE VII
IMPROVEMENT ANALYSIS

Comparison	**MAE Improvement**
Ensemble vs CNN (ResNet101)	30.45%
Ensemble vs Random Forest	45.24%
CNN vs MLP Baseline	96.11% (37×)

- **7-Day Recursive Forecast:** Multi-step prediction with drift analysis
- **Multi-Basin Generalisation:** Cross-validation across Pacific, Atlantic, Mediterranean
- **Per-Pixel Prediction:** Dense anomaly maps instead of scalar targets

TABLE VIII
DEVELOPMENT ITERATIONS

V.	**Change**	**MAE**	**Outcome**
v1	Basic CNN, 1 channel	0.076	No temporal context
v2	Huber loss	Worse	Forgives outliers
v3	RF on 63K pixels	0.126	Drowns in noise
v4	3-day RGB encoding	Better	Context added
v5	Oversampling top 10%	Helped	Spike awareness
v6	MSE + augmentation	0.083	Best individual
v7	50/50 ensemble (bad RF)	0.119	RF drags it down
v8	41 engineered features	0.128	RF now functional
v9	**Optimised ensemble**	**0.070**	**Final winner**

REFERENCES

[1] A. Parasyris et al., "Spatial forecasting of Marine Heatwave presence maps using Attention U-Net," *Mediterranean Sea Studies*, 2023.

[2] G. Bonino et al., "Machine learning methods to predict sea surface temperature and marine heatwave occurrence: a case study of the Mediterranean Sea," *Ocean Science*, vol. 20, pp. 417–432, 2024.

[3] N. Zrira et al., "Bidirectional LSTM with attention mechanism for Moroccan Sea SST forecasting," *J. Marine Systems*, 2023.

[4] A. Neto et al., "Hybrid residual modeling combining SVR and LSTM for Atlantic SST prediction," *Applied Ocean Research*, 2023.

[5] J. Xie et al., "3D U-Net for spatiotemporal volumetric learning in the South China Sea," *Remote Sensing*, 2024.

[6] J. X. Prochaska, P. C. Cornillon, and D. M. Reiman, "Deep learning of sea surface temperature patterns to identify ocean extremes," *Remote Sensing*, vol. 13, no. 4, p. 744, 2021.

[7] T. Song et al., "STL decomposition combined with SAM-LSTM for seasonal-trend analysis in China Seas," *Ocean Engineering*, 2023.

[8] K. Chen et al., "ConvTrans-CL: Causal convolution with Transformer and contrastive learning for global SST forecasting," *IEEE Trans. Geosci. Remote Sens.*, 2024.

[9] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE CVPR*, 2016, pp. 770–778.

[10] J. Taylor and M. Feng, "A deep learning model for forecasting global monthly mean sea surface temperature anomalies," *Frontiers in Climate*, vol. 4, p. 932932, 2022.

[11] J. Ren et al., "Prediction of sea surface temperature using U-Net based model," *Remote Sensing*, vol. 16, no. 7, p. 1205, 2024.

[12] A. Rauniyar et al., "Multi-dilated convolutional LSTM with U-Net for global sea surface temperature forecasting," *IEEE J. Sel. Topics Appl. Earth Obs. Remote Sens.*, 2024.

[13] H. Dai et al., "Sea surface temperature prediction by stacked generalization ensemble of deep learning," *Deep-Sea Research Part I*, vol. 211, p. 104370, 2024.

[14] C. Kumar et al., "A machine learning approach to estimating the error in satellite sea surface temperature retrievals," *Remote Sens. Environ.*, vol. 255, p. 112227, 2021.

[15] J. M. Lyman and G. C. Johnson, "Global high-resolution random forest regression maps of ocean heat content anomalies," *J. Atmos. Oceanic Technol.*, vol. 40, no. 5, pp. 575–586, 2023.

[16] P. Domel et al., "Event recognition in marine seismological data using Random Forest," *Geophysical J. Int.*, vol. 235, no. 1, pp. 589–609, 2023.

[17] E. Orlova et al., "Beyond ensemble averages: leveraging climate model ensembles for subseasonal forecasting," *Artificial Intelligence for the Earth Systems*, vol. 3, no. 4, 2024.

[18] B. Qiao et al., "Effective ensemble learning approach for SST field prediction using attention-based PredRNN," *Frontiers of Computer Science*, vol. 17, p. 171601, 2023.

[19] X. Bai et al., "Multivariate temperature prediction model based on CNN-BiLSTM and RandomForest," *J. Supercomputing*, vol. 81, p. 162, 2025.

[20] Y. Zhang, J. Liu, and W. Shen, "A review of ensemble learning algorithms used in remote sensing applications," *Applied Sciences*, vol. 12, no. 17, p. 8654, 2022.

[21] D. Marmanis et al., "Deep learning earth observation classification using ImageNet pretrained networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 1, pp. 105–109, 2016.

[22] F. Hu et al., "Transferring deep convolutional neural networks for the scene classification of high-resolution remote sensing imagery," *Remote Sensing*, vol. 7, no. 11, pp. 14680–14707, 2015.

[23] K. Nogueira, O.A.B. Penatti, and J.A. dos Santos, "Towards better exploiting convolutional neural networks for remote sensing scene classification," *Pattern Recognition*, vol. 61, pp. 539–556, 2017.

[24] Y. Ma et al., "Transfer learning in environmental remote sensing," *Remote Sens. Environ.*, vol. 301, p. 113924, 2024.

Fig. 6. Ensemble performance: (a) Prediction vs Ground Truth, (b) MAE comparison, (c) Predicted vs Actual scatter, (d) Error distribution.

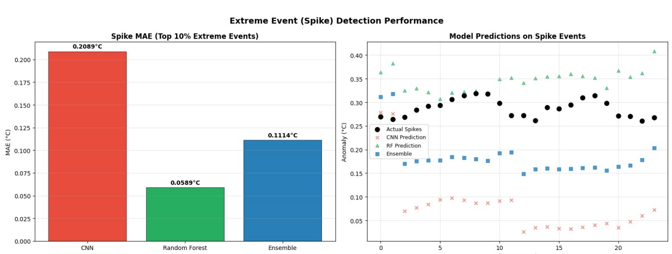

Fig. 7. Spike detection: CNN fails (0.2089) while RF excels (0.0589). Per-event predictions on top 10% extremes.

[25] R. B. Cleveland et al., "STL: A seasonal-trend decomposition procedure based on Loess," *J. Official Statistics*, vol. 6, no. 1, pp. 3–33, 1990.

[26] I. Corley et al., "Revisiting pre-trained remote sensing model benchmarks: resizing and normalization matters," in *Proc. IEEE/CVF CVPR Workshops*, 2024, pp. 3162–3172.

[27] F. Wang et al., "Deep learning for daily precipitation and temperature downscaling," *Water Resources Research*, vol. 57, no. 4, e2020WR029308, 2021.

[28] G. Douzas et al., "Imbalanced learning in land cover classification: improving minority classes' prediction accuracy using Geometric SMOTE," *Remote Sensing*, vol. 11, no. 24, p. 3040, 2019.

[29] A. Rauniyar et al., "Machine learning techniques for marine heatwave prediction: a comprehensive review," *Intelligent Marine Technology and Systems*, Springer, 2025.

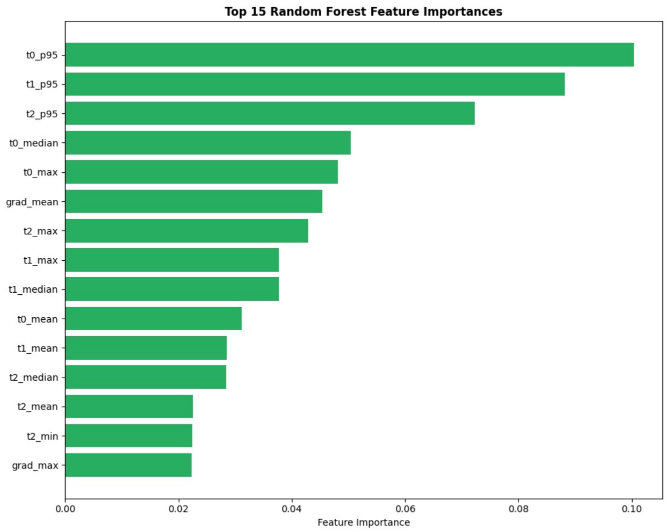

Fig. 8. Top 15 RF feature importances. The 95th percentile features dominate, capturing extreme temperature tails.

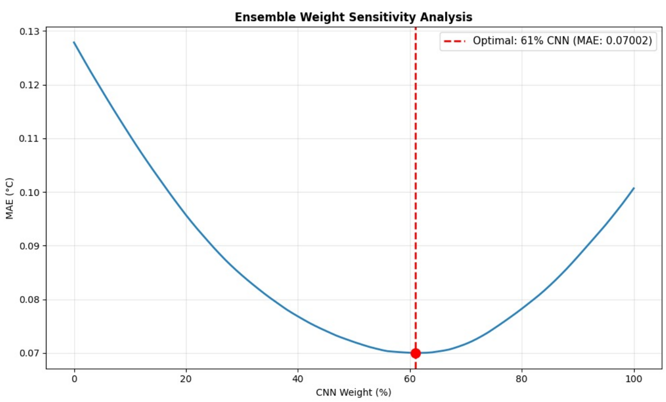

Fig. 9. Ensemble weight sensitivity: convex minimum at 61% CNN confirms complementarity.

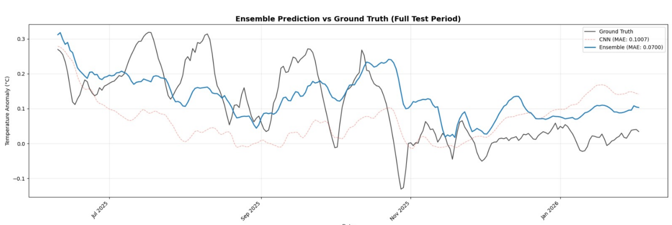

Fig. 10. Ensemble prediction vs ground truth over full test period (June 2025 – February 2026).

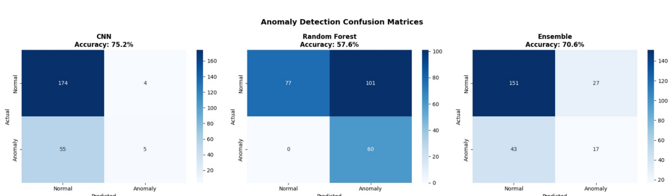

Fig. 11. Confusion matrices: CNN (75.2%), RF (57.6%), Ensemble (70.6%).

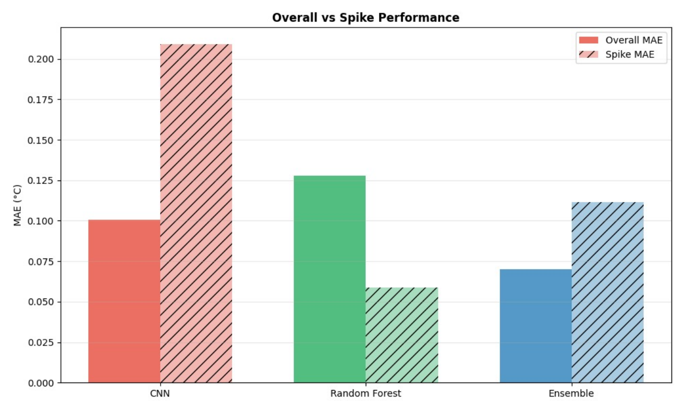

Fig. 12. Overall MAE vs Spike MAE demonstrating CNN-RF complementarity.